

FLOPs

1) Tensor shapes, FLOPs

$$X \in \mathbb{R}^{B \times 3 \times s' \times s'}$$

where B - batch size, s - padding

Q - Naïve

$$H_{out} = \left\lfloor \frac{H + 2P - k}{s} \right\rfloor + 1$$

- k - kernel size
 - s - Strider
 - P - Padding
-

upbeat Conv: $k=7, s=2, p=3$

$$H_{out} = \left\lfloor \frac{8+6-7}{2} \right\rfloor + 1 = \frac{9}{2}$$



$$B \times 3 \times S \times S \quad (S \text{ length kernel } 16)$$

$$\text{Conv } 7, S_2 \Rightarrow B \times 32 \times \frac{S}{2} \times \frac{S}{2}$$

$$\text{Pool}, S_2 \Rightarrow B \times 32 \times \frac{S}{4} \times \frac{S}{4}$$

$$\text{Conv } 5 \Rightarrow B \times 64 \times \frac{S}{4} \times \frac{S}{4}$$

$$\text{Conv } 3, S_2 \Rightarrow B \cdot 128 \cdot \frac{S}{8} \cdot \frac{S}{8}$$

$$\text{Conv } 1 \Rightarrow B \cdot 256 \cdot \frac{S}{8} \cdot \frac{S}{8}$$

$$\text{Conv } 3, S_2 \Rightarrow B \cdot 256 \cdot \frac{S}{16} \cdot \frac{S}{16}$$

$$\text{Conv } 1 \Rightarrow B \cdot 512 \cdot \frac{S}{16} \cdot \frac{S}{16}$$

$$\text{Global Avg Pool} \Rightarrow B \cdot 512$$

$$\text{Linear } : 512 \Rightarrow 256 \Rightarrow 100$$

2) FLOPS

$$1 \text{ MAC} = 2 \text{ FLOPs}$$

$$P_{\text{conv}} = 2 H_{\text{out}} W_{\text{out}} C_{\text{in}} C_{\text{out}} k^2$$

Для первого слоя:

$$P_1 = 2B \left(\frac{C}{2}\right)^2 \cdot 3 \cdot 32 \cdot 7^2$$

$$P_1 = 2352 \text{ BS}^2$$

Однмерная convolution:

$$P_2 = 6400 \text{ BS}^2$$

$$P_3 = 2304 \text{ BS}^2$$

$$P_4 = 1024 \text{ BS}^2$$

$$P_5 = 4608 \text{ BS}^2$$

$$P_6 = 1024 \text{ BS}^2$$

$$P_{\text{conv}} = 17712 \text{ BS}^2$$

$$P_{\text{Linear}} = 2B C_{\text{in}} C_{\text{out}}$$

$$\text{Linear}_1 = 2B \cdot 612 \cdot 256 = 262144 \text{ B}$$

$$Linear_2 = 2B \cdot 256 \cdot 100 = 51200B$$

$$P_{Linear} = 313344B$$

$$FLOPs(S, B) = B(12712S^2 + 313344)$$

Peak memory

FP32: 1 value 54 bytes

Forward: `lval() + inference_mode()`

$$N_{params} = 1040324$$

$$M_{params} = 1040324 \cdot 4$$

$$M_{params} = 4161296 \text{ bytes}$$

Input:

$$B \times 32 \times S$$

$$N_{input} = 3BS^2$$

Output:

$$B \times 32 \times \frac{S}{2} \times \frac{S}{2}$$

$$N_{\text{output}} = 32 B \frac{S^2}{4} = 8 B S^2$$

$$3 B S^2 + 8 B S^2 = 11 B S^2$$

$$N_{\text{activations}} = 11 B S^2 \cdot 4 = 44 B S^2$$

$$\text{Memory}(S, B) = 4161296 + 44 B S^2$$

Bytes moved

Bytes = input read + weights read + output write

$$N_{\text{moved}} = 1040324 + B(91 S^2 + 2448)$$

$$\text{Bytes}(S, B) = 4[1040324 + B(91 S^2 + 2448)]$$

Latency

$$P = \text{FLOPs}(S, B)$$

$$Q = \text{Bytes}(S, B)$$

$$T_{\text{memory}} = \frac{Q}{\beta}$$

note: β = effective memory bandwidth

$$\tau_{\text{compute}} = \frac{F}{\pi}$$

где π - effective compute throughput

t_0 - фиксированный overhead

который зависит: Latency $(S, B, \theta) = t_0 + \max\left(\frac{Q}{\beta}, \frac{F}{\pi}\right)$

где $\theta = (t_0, \beta, \pi)$

Результаты:

t_0 dominates $\Rightarrow$ launch-bound

$\frac{Q}{\beta}$ dominates $\Rightarrow$ memory-bound

$\frac{F}{\pi}$ dominates $\Rightarrow$ compute-bound

Energy

$$E = P \cdot t$$

Для количественной модели

$$P_G = \frac{\text{FLOPs}(S, B)}{10^9}$$

$$Q_G = \frac{\text{Bytes}(S, B)}{10^9}$$

$$\text{Energy}(S, B, \theta_E) = c_0 + c_f f_b + c_m Q_b$$

1gl:

$$\theta_E = (c_0, c_f, c_m)$$

Validation, Calibration

Beim Lernen gewohnter Teil: 75% $\Rightarrow$ Calibration

25% $\Rightarrow$ Validation

Parameter: θ, θ_E

we need some calibration points,

Validation points we use for fitting

There are also experiments:

Predicted Measured

we use validation points